

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DISCRETE MATHEMATICAL STRUCTURES [MAT 2101]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) In the divisibility order on $D_{12} = \{1, 2, 3, 4, 6, 12\}$, what is the least element? (1)
(A) 0
(B) 2
(C) 1
(D) 12
- 2) In the divisibility order on $D_{12} = \{1, 2, 3, 4, 6, 12\}$, what is the greatest element? (1)
(A) 1
(B) 12
(C) 6
(D) 4
- 3) In the divisor lattice D_{12} , what is $4 \wedge 6$? (1)
(A) 2
(B) 4
(C) 6
(D) 12
- 4) In the divisor lattice D_{12} , what is $4 \vee 6$? (1)
(A) 2
(B) 4
(C) 6
(D) 12
- 5) Which is a complement of 2 in D_{12} , with meet given by gcd and join by lcm? (1)
(A) 3
(B) 6
(C) No element
(D) 12

Section B: Theory questions (25 marks)

- 6) For $F(x, y, z) = (x \wedge \neg y) \vee z$, construct the truth table and write the canonical disjunctive normal form and canonical conjunctive normal form. Use variable order (x, y, z) . (3)
- 7) Prove the Boolean identity $a \vee (\neg a \wedge b) = a \vee b$ using distributive and complement laws. Name the laws used. (2)
- 8) Count the integer solutions of $x + y + z = 12$ subject to $1 \leq x \leq 5$, $2 \leq y \leq 6$ and $z \geq 0$. Transform to nonnegative variables and use inclusion-exclusion. Show each excluded case. (5)
- 9) Find the 42nd lexicographic permutation of the symbols 0, 1, 2, 3, 4, with 01234 counted as the first. Show the factorial-block choices. (3)
- 10) How many distinct arrangements of the letters of BALLOON are possible? How many have the two Ls adjacent? Explain how repeated letters are handled. (2)
- 11) Use Dijkstra's algorithm from A in an undirected weighted graph with vertices A, B, C, D, E and edges AB:4, AC:1, BC:2, BD:1, CD:5, CE:8, DE:3. Show the tentative-distance table after settling each vertex. Give a shortest path and distance from A to every other vertex. (5)
- 12) Prove that a simple graph with $n \geq 2$ vertices and minimum degree at least $(n - 1)/2$ is connected. Hint: bound the size of each component if the graph were disconnected. (3)
- 13) A simple graph is regular and self-complementary on n vertices. Use degree counting to show its degree is $(n - 1)/2$ and then show that $n \equiv 1 \pmod{4}$. (2)

End of question paper